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Inventor(s)	Dubois; Abigail L. et al.

Hair clip having a hidden hinge

Abstract

A hair clip having a hidden hinge includes two clip members each having a handle, an array of teeth for retaining hair, an engagement shoulder disposed between the handle and the array of teeth, and a resilient member disposed along the inner edges of the engagement shoulders and between the arrays of teeth of each clip member interconnecting and biasing the clip members toward a closed configuration, the engagement shoulders disposed in abutting engagement thereby forming an engagement seam, and the engagement shoulders disposed between the handles and the resilient member of the two clip members such that the resilient member is hidden behind the engagement seam when viewing the hair clip from an outer vantage point.

Inventors: **Dubois; Abigail L. (Danbury, CT), Dubois; Lawrence H. (Danbury, CT)**

Applicant: **Each Jewels LLC** (New York, NY)

Family ID: **94483042**

Assignee: **Each Jewels LLC (New York, NY)**

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
884680	12/1907	Polakoskey	211/119.13	A41B 3/08
2459926	12/1948	Benedetto	132/277	A45D 8/20
2540042	12/1950	Craigen	132/277	A45D 8/20
3092870	12/1962	Baer	N/A	N/A
3341242	12/1966	Carson	D7/706	A47G 23/0208
3369585	12/1967	Martinsen	150/147	A45C 11/182
D210860	12/1967	Franciscus	D28/40	N/A
5337451	12/1993	Goossens	N/A	N/A
5787905	12/1997	Yasuda	132/277	A45D 8/20
5842482	12/1997	Potut	N/A	N/A
5873377	12/1998	Yang	132/273	A45D 8/20
6490768	12/2001	Goodall	24/339	A47C 21/022
2003/0094555	12/2002	Prizzi	248/214	G09F 7/18
2009/0272397	12/2008	Defenbaugh et al.	N/A	N/A
2009/0277469	12/2008	Hsu	132/277	A45D 8/20
2010/0242987	12/2009	Washington	132/277	A45D 8/20
2021/0022469	12/2020	Freeman	N/A	A45D 8/20

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
200306880	12/2002	KR	N/A

Primary Examiner: Lucchesi; Nicholas D

Assistant Examiner: Gill; Jennifer

Attorney, Agent or Firm: Beeson Skinner Beverly, LLP

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of application Ser. No. 18/594,927, filed Mar. 4, 2024, which is a continuation-in-part of application Ser. No. 18/375,962, filed Oct. 2, 2023, which is a continuation-in-part of application Ser. No. 18/231,744, filed Aug. 8, 2023.

BACKGROUND

Technical Field

(1) This invention relates to hair clips and more particularly to hair clips having a hinge mechanism positioned around and hidden behind inwardly extending engagement shoulders of the handles of the hair clip which form an engagement seam around which the clip members pivot.

Description of Related Art

(2) Prior to the invention disclosed herein, a typical hair clip comprises two or more clip members coupled together by an outwardly visible spring-biased hinge. This hinge can have a central pivotal axis, such as in a traditional claw clip, or a side pivotal axis, such as in a side claw clip, alligator clip, or salon clip or spring steel snap clip. Retaining structures, aka, knuckles, and springs are generally visible, creating an undesirable appearance, as well as opportunities to break, tangle, or damage hair.

SUMMARY OF THE INVENTION

(3) The invention discloses a hair clip with a resilient hinge mechanism such as a spring. The spring mechanism is disposed on the interior side of and engagement seam formed by two inwardly extending engagement shoulders thus presenting from an outer vantage point a single engagement seam.

(4) As in a traditional hair clip, the clip members include one or more levers or handles used to widen the space between the teeth or combs. Placing the hinge mechanism between the teeth in the interior of the hair clip allows for a wider opening than with a traditional hair clip. When the handles are released, the hair is securely held between the teeth.

(5) There are also aesthetic benefits to the invention: the hinge mechanism faces inwards towards the retained hair. Thus, hinge knuckles, hinge pins, and/or springs are partially or completely covered and not outwardly visible. An optional spring cover can be employed to prevent hair contact with the spring mechanism, but it is not a requirement.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1**A is an upper perspective view of a first embodiment of a hair clip having a hidden hinge shown in a closed configuration.

(2) FIG. **1**B is a bottom perspective view thereof showing the hair clip in a partially open configuration.

(3) FIG. **1**C is a sectional view thereof taken along lines **1**C-**1**C of FIG. **1**B.

(4) FIG. **1**D is a bottom perspective view thereof showing the hair clip in a fully open configuration.

(5) FIG. **1**E is a side elevational view thereof showing the hair clip in a fully open configuration.

(6) FIG. **1**F is an enlarged portion of a portion of another embodiment of the invention similar to that shown in FIGS. **1**A-**1**E.

(7) FIGS. **2**A-**2**D show representative examples of how the first embodiment of the invention disclosed herein may be applied to four different hair claw clip styles.

(8) FIG. **3**A is an upper right rear perspective view of a second embodiment thereof showing the hair clip in a closed configuration.

(9) FIG. **3**B is an upper right front perspective view thereof showing the hair clip in an open

configuration.
(10) FIG. **3**C is an enlarged perspective view of a portion of FIG. **3**B showing the hinge mechanism thereof.
(11) FIG. **3**D is front elevational view thereof showing the hair clip in an open configuration.
(12) FIG. **3**E is side elevational view thereof showing the hair clip in an open configuration.
(13) FIG. **3**F is an enlarged view of a portion of another embodiment of the invention similar to that shown in FIGS. **3**A-**3**E.
(14) FIGS. **4**A-**4**D show representative examples of how the second embodiment of the invention disclosed herein may be applied to four different side claw clip styles.

DETAILED DESCRIPTION OF THE ILLUSTRATED EMBODIMENT

(15) A first embodiment of hair clip having a hidden hinge, indicated at **100** in FIGS. **1**A-**1**E, comprises two adjacent clip members **110** each including a grabbable handle **115** and an array of teeth **120**. Each handle **115** includes an abutment face **116** which faces the abutment face **116** of the other handle **115**, and an obverse face **117** on the opposite side of the handle **115**. Arrays of teeth **120** extend from the obverse faces **117** of the handles **115** between the outer portions **118** and the inner portions **119** thereof as shown. Inwardly facing engagement shoulders **121** (see FIGS. **1**B-**1**E) on the inner ends of the handles **115** are in abutting disposition and form between them an engagement seam **125** extending along the arrays of teeth **120** of the clip members **110**. While the engagement seam **125** in the illustrated embodiment is shown extending continuously along the full lengths of the handles **115** and teeth **120**, in other embodiments gaps can be provided in or along the engagement seam so that it is separated into two or more engagement seams.
(16) The clip members **110** are coupled and held together by a spring **145** or other resilient material that extends across and is adjacent to the inner edges **122** of engagement shoulders **121**. In this application the word “inner” and variants thereof refer generally to the area bounded by the arrays of teeth **120** and the engagement shoulders **121** of the clip members **110**. Thus, as can be seen from FIGS. **1**B-**1**E, the abutting engagement shoulders **121** form a functional hinge **130** about which the clip members **110** pivot, the handles **115** acting as levers that open the teeth **120**. And, when viewing the hair clip from an outer or handle-side vantage point, such as shown in FIG. **1**A, the spring **145** is hidden.
(17) With continuing reference to FIGS. **1**B-**1**E, pivot pins **135** disposed in retaining structures **140** create axes around which handles **115** and teeth **120** pivot. The clip members **110** are coupled together by dual axis torsion springs **145** which return the claw clip to its closed position in which the teeth **120** are positioned to retain hair. Each torsion spring **145** has two coiled portions **150** interconnected by connector **155**. Connectors **155** that comprise resilient or non-resilient materials are understood to be within the scope of this invention. One of the pivot pins **135** is rotatably received in the center bore formed by each coiled portion **150** such that the coiled portions **150** of each spring **145** wind around and extend along the axes formed by pivot pins **135** and are spaced apart perpendicularly to the axial extent thereof. Each coiled portion **150** is disposed on the inner portion **119** of the obverse face **117** of one of the handles **115** and is thus affixed to the teeth **120** of one of the clip members **110**, so that the coiled portions **150** are wound more tightly as the clip members pivot from the closed configuration, seen in FIG. **1**A, to an open configuration, such as seen in FIGS. **1**B-**1**E.
(18) In the illustrated embodiment, a short extension **160** of spring **145** is affixed behind catch **165** to the inner portion **119** of the obverse face **117** of each handle **115** and biases the clip members **110** toward the closed configuration. It will be understood, however, that the coiled portions **150** of the springs **145** may be attached directly to the teeth **120** of the clip members **110** in various ways that bias the clip members **110** toward a closed configuration.
(19) It will be understood that, while the retaining structures **140** and pivot pins provide pivot axes and anchor points for the dual axis torsion springs **145**, only the engagement shoulders **121** and one or more resilient structures such as springs are needed to create a functional hinge. In other

embodiments hinge types may include, but are not limited to, two-pin hinges, hinges with virtual pivot axes, single pin hinges, single pin hinges in combination with a virtual pivot axis and living hinges. Such hinges are biased toward the closed position using resilient or elastomeric materials or components or springs. While the embodiments described in this application focus on springs, other resilient materials or devices that have an elastic body that recovers its original shape when released after being distorted fall within the ambit of this invention.

(20) Positioning the engagement shoulders **121** between the handles **115** and the spring **145** of each clip member **110** hides all or part of the spring **145** behind the handles **115**, the engagement shoulders **121**, and the engagement seam **125** when the hair clip is viewed from an outer, or handle-side, vantage point as seen in FIG. **1**A. This provides a unique aesthetic advantage over prior art hinged hair claw clips and is particularly advantageous when the teeth **120** of the hair clip are in use retaining hair because only the handles **115** and the engagement seam **125** between the handles generally will then be visible.

(21) In other embodiments of the device a separate cover may be provided between the teeth of the clip members that cover and hide the spring mechanism from an inner vantage point. Thus, with reference to FIG. **1**F, in an enlarged view of a portion of another embodiment **101** of a hair clip according to the invention, similar to the embodiment shown in FIG. **1**E, the coiled portions **150** and a length of the pivot pins **135** (both indicated in broken lines) of the spring **145** are enclosed in retaining structures **141**. Each of the retaining structures **141** includes an abutment face **116** which engages the abutment face **116** of another of the retaining structures in the open position as shown.

(22) FIGS. **2**A-**2**D show additional examples of how the invention may be applied to four representative styles of hair claw clips having four different types of hinges. FIG. **2**A shows an ornamental hair claw clip with two clip members **200** and a single seamed hinge **210**, wherein the clip members **200** pivot about the pins **235** in the claw clip. Springs on the underside of the claw, which ensure closure, are not illustrated.

(23) FIG. **2**B shows an extruded ornamental hair claw clip having two clip members **220** and a single seamed hinge **230**, wherein the clip members **220** pivot about a hidden hinge on the underside of claw clip. The resilient material provided on the underside of the claw to ensure closure is not shown for clarity.

(24) FIG. **2**C is a perspective view of an extruded star ornamental hair claw clip having two clip members **240** and a single seamed hinge **250**, wherein the clip members **240** pivot about pivot pins **255** in the claw clip. An elastomer provided on the underside of the claw clip to ensure closure is not shown for clarity.

(25) FIG. **2**D is a perspective view of a heart shaped ornamental hair claw clip with two clip members **260** and a single seamed hinge **270**, wherein the clip members **260** pivot about a hidden hinge in the claw clip. Springs provided on the underside of the claw to ensure closure are not shown for clarity. It is understood that one or more body members may be made from resilient material to open and close the clip members to readily retain hair.

(26) FIGS. **3**A-**3**E show a second embodiment of the disclosed invention in the form of a side hair claw clip. The hair clip is comprised of two adjacent clip members **310** each having a handle **315** and an array of teeth **320**. As in the first embodiment of the invention discussed above in connection with FIGS. **1**A-**1**D, only a singular engagement seam **325** is visible while in use in a closed configuration as best seen in FIG. **3**A. Engagement shoulders **330** protruding inwardly from the base portions **335** of the teeth **320** of each clip member **310** are disposed in abutting engagement thereby providing a rolling pivot axis around which the clip members **310** pivot and form the engagement seam **325**. An extension spring **340** is disposed in opposing recesses **345** provided in shoulders **330** as seen in FIGS. **3**B-**3**D. In the embodiment shown in FIGS. **3**B-**3**C each end of the extension spring **340** is fixed to one of the engagement shoulders **330** via anchor pins **350** such that the spring **340** joins the clip members **310** together while biasing them toward the closed configuration shown in FIG. **3**A. As seen in FIG. **3**D, each end **342** of the spring **340** is

embedded directly into the base portion **335** of the teeth **320** of one of the clip members **310**.
(27) In the embodiments illustrated in FIGS. **3**A-**3**E handles **315** act as levers that open the teeth **320** of the side hair claw clip members **310** while springs **320** urge the teeth **320** of the clip to the closed configuration such as for retaining hair. It is understood that pivot pins or spring hinge mechanisms are not required for this invention to function. Body members or coupled springs may comprise resilient or elastomeric materials used to bias body members and retain hair.
(28) With reference now to FIG. **3**F, in an enlarged view of a portion of another embodiment **301** of a hair clip according to the invention, similar to the embodiment shown in FIG. **3**D, anchor pins **355** and spring **340** (both indicated in broken lines) are enclosed in engagement shoulders **330** so that they are hidden when the hair clip is viewed from an internal vantage point.
(29) The invention disclosed herein is applicable to side hair claw clips of all hinge styles. FIGS. **4**A-**4**D show additional examples of how the invention may be applied to four representative styles of side hair claw clips with four different types of hinges.
(30) FIG. **4**A is a perspective view of an extruded ornamental side hair claw clip having two clip members **400** which pivot about a single seamed hinge **410**. A spring provided on the underside of the clip to ensure closure is not shown for clarity. It is understood that one or more body members may be made from resilient material to open and close the clip members to readily retain hair.
(31) FIG. **4**B is a perspective view of a seashell ornamental side hair claw clip having two clip members **420** which pivot about a single seamed hinge **430**. The resilient material provided on the underside of the claw to ensure closure is not shown for clarity.
(32) FIG. **4**C is a perspective view of a flower and leaf ornamental side hair claw clip having two clip members **440** which pivot about a single seamed hinge **450**. A resilient component on the underside of the clip that both connects and aligns the two clip members is not illustrated.
(33) FIG. **4**D is a perspective view of an ornamental side hair claw clip having two clip members **460** and a single seamed hinge **470**, which rotates along the pins **480** in the claw clip. Springs provided on the underside of the claw to ensure closure are not shown for clarity. It is understood that one or more body members may be made from resilient materials to open and close the clip members to readily retain hair.
(34) While two pin and virtual hinges are used in most of the examples shown above, one skilled in the art will understand that these are just for illustrative purposes and all types of hinges including, but not limited to, living hinges, single pin hinges, piano hinges, geared hinges, etc. may be employed as long as there is a single outermost seam. Similarly, resilient materials of all types, e.g., elastomeric materials, spring-metals, all manner of springs, e.g., torsion springs, coupled torsion springs, extension springs, leaf springs, clock springs, and components, such as split rings, may be used to bias the hair clip to a closed configuration.
(35) While no hinge pins, knuckles, or connectors are required for the hinge to operate effectively, they are not precluded from use in this invention. Furthermore, independent of the type of resilient materials, components, and/or springs used to connect and orient the leaves, the seam between the two leaves does not have to be a single straight line but can be of any arbitrary shape depending on the aesthetics desired. Thus, it is understood that variations in the shape of the gap including, but not limited to, polygonal or curved shapes, e.g., portions of circles, parabolas, ellipses, catenaries, or organic shapes, fall within the ambit of this invention. Furthermore, the cross-section of the gap may be different on each leaf.
(36) There have thus been described and illustrated certain embodiments of spring hair clips with a single outermost seam according to the invention. Although the present invention has been described and illustrated in detail, it should be clearly understood that the disclosure is illustrative only and is not to be taken as limiting, the spirit and scope of the invention being limited only by the terms of the appended claims and their legal equivalents.

Claims

1. A hair clip having a hidden hinge, the hair clip comprising: a first and a second clip member, both of the first and second clip members having a grabbable handle, an array of teeth extending away from the handle, and one or more inwardly extending engagement shoulders disposed between the handle and the array of teeth, at least one of said one or more engagement shoulders of said first clip member in abutting engagement with one of the one or more engagement shoulders of said second clip member along a continuous linear engagement seam, said engagement seam extending substantially parallel with the array of teeth of at least one of said first and second clip members, so that said first and second clip members pivot about said engagement seam between open and closed configurations, and one or more resilient members coupling together said first and second clip members, at least one of the one or more engagement shoulders of said first and second clip members disposed between the handles and said one or more resilient members of said clip members, so that all portions of the one or more resilient members are hidden when viewing the hair clip from an outer vantage point, such that the one or more resilient members bias said first and second clip members from the open configuration toward the closed configuration and retain at least one of the one or more engagement shoulders of said first member in continuous abutting engagement with at least one of the one or more engagement shoulders of said second clip member when pivoting said first and second clip members between the open and closed configurations.
2. The hair clip of claim 1, further comprising: each of the one or more engagement shoulders of said first and second clip members having an inner edge disposed between the array of teeth and the engagement seam, and one of said one or more resilient members disposed adjacent the inner edges of the one or more engagement shoulders of said first and second clip members.
3. The hair clip of claim 1, wherein the one or more resilient members comprise one or more springs.
4. The hair clip of claim 1, wherein said engagement shoulders of one or both of said first and second clip members have an arcuate shape along the entire length of said engagement seam.
5. The hair clip of claim 4 wherein said engagement shoulders of one or both of said first and second clip members have said arcuate shape along the entire length of said engagement seam.
6. The hair clip of claim 1, wherein each of the one or more resilient members has a tensed state and a relatively relaxed state, each of said first and second clip members has a grabbable handle, in the closed configuration, the handles of said first and second clip members are spaced apart, the one or more resilient members are in the relatively relaxed state, and the arrays of teeth of said first and second clip members are in adjacent and overlapping position, and in the open configuration, the handles of said first and second clip members are generally in adjacent disposition, the one or more resilient members are in the tensed state, and the arrays of teeth of said first and second clip members are spaced apart.
7. The hair clip of claim 1, wherein said engagement seam is disposed between the handles of said first and second clip members.
8. The hair clip of claim 1, wherein each handle of said first and second clip members has an abutment face facing the abutment face of the other of said first and second clip members, each handle having an obverse face on the opposite side thereof, at least one end of at least one of said one or more resilient members including a coiled portion disposed on the obverse face of the handle between the engagement seam and the array of teeth, and in the open configuration said coiled portion is more tightly wound than when in the closed configuration.
9. The hair clip of claim 8, wherein said at least one end comprises at least two ends including first and second ends, each of said first and second ends having a coiled portion affixed to one of said first and second clip members, each of said coiled portions having an axial extent, and the coiled portions of said first and second ends spaced apart perpendicularly to the axial extents thereof, and

at least one of said one or more resilient members includes a connector extending between and integrally coupling the coiled portions of said first and second ends, each of the coiled portions of said first and second ends disposed on opposite sides of said engagement seam, in the open configuration the coiled portion of each of said first and second ends more tightly wound than when in the closed configuration.

10. The hair clip of claim 9, further comprising: the connector comprising resilient materials.

11. The hair clip of claim 9, further comprising: the coiled portions of said first and second clip members defining a center opening having an axial dimension, said connector forming a loop extending perpendicularly to the axial dimension of said coiled portions to accommodate said engagement shoulders as said first and second clip members move between the open and closed configurations.

12. The hair clip of claim 8 further comprising: at least one of said first and second clip members having one or more retaining structures, each of the one or more retaining structures having an axially extending bore, each of the coiled portions of the ends of the one or more resilient members defining a center opening, and one or more pivot pins, one of the one or more pivot pins received in the bore of at least one of the one or more retaining structures of one of said clip members and received in the center opening of the coiled portion of one end of at least one of the resilient members, such that said pivot pin defines a pivot axis about which one of said clip members pivots.

13. The hair clip of claim 12 wherein each of the one or more retaining structures forms a portion of one of the one or more engagement shoulders.

14. The hair clip of claim 12 further comprising: at least one of the one or more retaining structures of both of said first and second clip members having an abutment face, and in the open configuration the abutment face of the at least one of the one or more retaining structures of one of said first and second clip members abutting the abutment face of the at least one of the one or more retaining structures of the other of said first and second clip members, thereby forming a portion of said engagement seam.

15. The hair clip of claim 12 further comprising: each of the two clip members having one or more retaining structures, each of the coiled portions of the ends of the one or more resilient members defining a center opening, said one or more pivot pins comprising two or more pivot pins, and said two or more pivot pins in parallel disposition, each of the two or more pivot pins having an axial dimension, each one of the two pivot pins received in at least one of the one or more retaining structures of one of said first and second clip members and in the center opening of the coiled portion of one end of at least one of the resilient members, the pivot pin received in one of said at least one or more retaining structures of one of said first and second clip members spaced apart perpendicularly to the axial dimension of the pivot pin received in at least one of said one or more retaining structures of the other of said first and second clip members.

16. The hair clip of claim 15, wherein each pivot pin defines a pivot axis about which one of said first and second clip members pivots, the pivot axis of each of said first and second clip members spaced apart perpendicularly from the pivot axis of the other of said first and second clip members.

17. The hair clip of claim 8, further comprising: said at least one end of at least one of the one or more resilient members comprising at least two ends including first and second ends said first and second ends each having a coiled portion, the coiled portions of said first and second ends connected by an uncoiled connector.

18. A hair clip having a hidden hinge, the hair clip comprising: first and second clip members, each of said clip members having a grabbable handle, an array of teeth extending away from the handle, and one or more inwardly extending engagement shoulders, at least one of the one or more engagement shoulders of each of said first and second clip members in abutting engagement with one of the one or more engagement shoulders of the other of said first and second clip members along a continuous linear engagement seam extending substantially parallel with the array of teeth of at least one of said first and second clip members and about which said clip members pivot

between open and closed configurations, and one or more resilient members coupling together said first and second clip members, each of the one or more resilient members having two ends, one or both of the two ends of at least one of the one or more resilient members including a coiled portion affixed to one of said first and second clip members, each of the coiled portions more tightly wound when said first and second clip members are in the open configuration than when in the closed configuration, at least one of the one or more engagement shoulders disposed between the handles of said first and second clip members and the one or more resilient members, so that all of the one of the one or more resilient members is hidden when viewing the hair clip from an outer vantage point when said first and second clip members are in the closed configuration or in the open configuration, such that the one or more resilient members bias said first and second clip members from the open configuration to the closed configuration and retain the engagement shoulders of said first and second clip members in continuous abutting engagement during pivoting motion of the handles between the open and closed configurations.

19. The hair clip of claim 18, wherein the one or more resilient members comprises at least one spring.